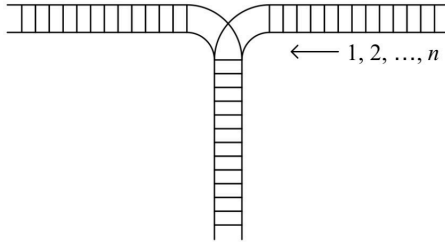


1. (10%) Consider the railroad switching network. Railroad cars numbered $0, 1, 2, \dots, n-1$ are at the right. Each car is brought into the stack and removed at any time. For instance, if $n = 3$, we could move in 0 , move in 1 , move in 2 , and then take the cars out, producing the new order $2, 1, 0$. Find all possible permutations for $n = 4$.



Ans: total possible permutations is 14.
my code and answer is below.

```
18 inline bool isValid( const vi check ){
19     bool isPossible = 1;
20     stack<int> st;
21     for ( int i = 0 , j = 0 ; i < 4 ; i++ ){
22         while( j <= check[i] ){
23             st.push(j);
24             ++j;
25         }
26
27         if ( !st.empty() ){
28             if ( st.top() == check[i] ) st.pop();
29             else{
30                 isPossible = 0;
31                 break;
32             }
33         }
34
35         if ( j > 4 and st.empty() ) break;
36     }
37     return isPossible;
38 }
39
40 signed main(){
41     whitebear;
42     vi train{0, 1, 2, 3};
43     int cnt = 1;
44     cout << train << "\n";
45     while ( next_permutation(train.begin(), train.end()) ){
46         if ( isValid(train) ){
47             cout << train << "\n";
48             ++cnt;
49         }
50     }
51     cout << "Total Valid Count:" << cnt << "\n";
52     return 0;
53 }
```

```
同画 输出 连接埠 终端机 侦错主控台
C++ d:\Project\C++\NT0U-DS\hw2_problem1.exe
0 1 2 3
0 1 3 2
0 2 1 3
0 2 3 1
0 3 2 1
1 0 2 3
1 0 3 2
1 2 0 3
1 2 3 0
1 3 2 0
2 1 0 3
2 1 3 0
2 3 1 0
3 2 1 0
Total Valid Count:14
```

2. (20%) Given a **stack** of size 4, according to Program 3.1 to 3.3 in the textbook, what is the stack in each step when we do the following commands:

push J1, push J2, push J3, pop, push J4, push J5, push J6

top	Q[0]	Q[1]	Q[2]	Q[3]	Comments
-1					stack is empty
0	J1				J1 added
1	J1	J2			J2 added
2	J1	J2	J3		J3 added
1	J1	J2			J3 deleted
2	J1	J2	J4		J4 added
3	J1	J2	J4	J5	J5 added
3	J1	J2	J4	J5	Stack is full

3. (20%) Given a **queue** of size 4, according to Program 3.5 to 3.6 in the textbook, what is the stack in each step when we do the following commands:

add J1, add J2, add J3, delete, add J4, add J5, add J6

front	rear	Q[0]	Q[1]	Q[2]	Q[3]	Comments
-1	-1					queue is empty
-1	0	J1				J1 added
-1	1	J1	J2			J2 added
-1	2	J1	J2	J3		J3 added
0	2		J2	J3		J1 deleted
0	3		J2	J3	J4	J4 added
0	3		J2	J3	J4	Queue is full
0	3		J2	J3	J4	Queue is full

4. (20%) Given a **circular queue** of size 4, according to Program 3.7 to 3.8 in the textbook, what is the stack in each step when we do the following commands:

~~add J1, add J2, add J3, delete, add J4, add J5, delete, delete~~

front	rear	Q[0]	Q[1]	Q[2]	Q[3]	Comments
0	0					queue is empty
0	1		J1			J1 added
0	2		J1	J2		J2 added
0	3		J1	J2	J3	J3 added
1	3			J2	J3	J1 deleted
1	0	J4		J2	J3	J4 added
1	0	J4		J2	J3	Queue is full
2	0	J4			J3	J2 deleted

3 0 J4

J3 deleted

5. (20%) Write the postfix form of the following infix expressions:

(a) $(A + B) * C + D / (E + F * G) - H$

(b) $(A \&\& B) \parallel C \parallel !(E > F) * "!" > ">" < "<" > "&\&" > "||"$

(a)

$$\begin{aligned}
 &(((A+B)*C) + (D/(E+(F*G))))-H \\
 &\quad AB+C* \quad D \quad EFG*+ /+ H- \\
 &= AB+C*DEFG*+ /+ H-
 \end{aligned}$$

(b)

$$\begin{aligned}
 &(((A\&\&B)\parallel C)\parallel !(E>F)) \\
 &\quad AB\&\&C\parallel \quad EF>! \parallel \\
 &= AB\&\&C\parallel EF>!\parallel
 \end{aligned}$$

6. (10%) Write the infix form of the following postfix expression:

A B C < C D > || ! && ! C E < ||

Ans: $!(A \&\& !((B < C) || (C > D))) || (C < E) \neq$

~~D~~
~~C~~ ~~C~~ ~~C > D~~
~~B~~ ~~B < C~~ $!((B < C) || (C > D))$ ~~E~~
~~A~~ $!(A \&\& !((B < C) || (C > D)))$ $C < E$